

Almost Sure Convergence: Selected Proofs

MBZUAI: SDS8102 – Advanced Mathematical Statistics

Kolmogorov's criterion, Three-series theorem, SLLN

1. Kolmogorov's Inequality and Almost-Sure Convergence

We now develop a collection of tools for proving almost-sure convergence of sums of independent random variables. These results will also lead to a proof of the strong law of large numbers.

1.1. Kolmogorov's Maximal Inequality

Chebyshev's inequality controls the value of a partial sum at a fixed time. For almost-sure convergence, however, we often need to control the entire path of partial sums simultaneously.

Theorem 1.1 (Kolmogorov's Maximal Inequality). *Let $X_1, \dots, X_n$ be independent random variables satisfying*

$$\mathbb{E}[X_i] = 0, \quad \text{Var}(X_i) < \infty.$$

Define

$$S_k = \sum_{i=1}^k X_i.$$

Then, for every $\lambda > 0$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq \lambda\right) \leq \frac{\text{Var}(S_n)}{\lambda^2} = \frac{1}{\lambda^2} \sum_{i=1}^n \text{Var}(X_i).$$

Proof. Define the disjoint events

$$A_k = \{|S_j| < \lambda \text{ for } j < k, \quad |S_k| \geq \lambda\}.$$

Thus A_k is the event that the partial sums first cross the level λ at time k . Moreover,

$$\bigcup_{k=1}^n A_k = \left\{ \max_{1 \leq k \leq n} |S_k| \geq \lambda \right\}.$$

Write

$$S_n = S_k + (S_n - S_k).$$

Since A_k depends only on $X_1, \dots, X_k$, while $S_n - S_k$ depends only on $X_{k+1}, \dots, X_n$, independence and centering give

$$\mathbb{E}[S_k(S_n - S_k)\mathbf{1}_{A_k}] = 0.$$

Therefore

$$\begin{aligned} \mathbb{E}[S_n^2 \mathbf{1}_{A_k}] &= \mathbb{E}[S_k^2 \mathbf{1}_{A_k}] + \mathbb{E}[(S_n - S_k)^2 \mathbf{1}_{A_k}] \\ &\geq \mathbb{E}[S_k^2 \mathbf{1}_{A_k}] \\ &\geq \lambda^2 \mathbb{P}(A_k). \end{aligned}$$

Summing over k and using disjointness,

$$\mathbb{E}[S_n^2] \geq \sum_{k=1}^n \mathbb{E}[S_n^2 \mathbf{1}_{A_k}] \geq \lambda^2 \sum_{k=1}^n \mathbb{P}(A_k).$$

Hence

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq \lambda\right) \leq \frac{\mathbb{E}[S_n^2]}{\lambda^2}.$$

Since the variables are independent and centered,

$$\mathbb{E}[S_n^2] = \text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i).$$

■

Why this is stronger than Chebyshev. Chebyshev gives only

$$\mathbb{P}(|S_n| \geq \lambda) \leq \frac{\text{Var}(S_n)}{\lambda^2},$$

whereas Kolmogorov's inequality controls

$$\max_{1 \leq k \leq n} |S_k|$$

with exactly the same upper bound.

1.2. Reverse Kolmogorov Maximal Inequality

The following reverse form of Kolmogorov's maximal inequality is useful when the summands are uniformly bounded.

Theorem 1.2 (Reverse Kolmogorov Maximal Inequality). *Let $X_1, \dots, X_n$ be independent random variables satisfying*

$$\mathbb{E}[X_k] = 0, \quad |X_k| \leq c \quad a.s.$$

Define

$$S_k = \sum_{j=1}^k X_j, \quad V_n = \sum_{j=1}^n \text{Var}(X_j).$$

Then, for every $\lambda > 0$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \leq \lambda\right) \leq \frac{(\lambda + c)^2}{V_n}.$$

Equivalently,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| > \lambda\right) \geq 1 - \frac{(\lambda + c)^2}{V_n}.$$

Proof. Define the stopping time

$$\tau = \min \{k \leq n : |S_k| > \lambda\},$$

with the convention $\tau = n$ if no crossing occurs.

Consider the stopped sum

$$S_\tau = \sum_{j=1}^n X_j \mathbf{1}_{\{\tau \geq j\}}.$$

Since $\{\tau \geq j\}$ depends only on $X_1, \dots, X_{j-1}$, it is independent of X_j . Moreover, the cross terms vanish by independence and centering. Hence

$$\mathbb{E}[S_\tau^2] = \sum_{j=1}^n \mathbb{E}[X_j^2] \mathbb{P}(\tau \geq j) = \sum_{j=1}^n \text{Var}(X_j) \mathbb{P}(\tau \geq j).$$

Let

$$A = \left\{ \max_{1 \leq k \leq n} |S_k| \leq \lambda \right\}.$$

On A , no crossing occurs, and therefore $\tau \geq j$ for every $j \leq n$. Consequently,

$$\mathbb{P}(\tau \geq j) \geq \mathbb{P}(A),$$

and hence

$$\mathbb{E}[S_\tau^2] \geq \mathbb{P}(A) \sum_{j=1}^n \text{Var}(X_j) = V_n \mathbb{P}(A).$$

On the other hand, at the first crossing the overshoot is at most c . Therefore

$$|S_\tau| \leq \lambda + c$$

almost surely, and thus

$$\mathbb{E}[S_\tau^2] \leq (\lambda + c)^2.$$

Combining the two bounds gives

$$V_n \mathbb{P}(A) \leq (\lambda + c)^2.$$

Therefore

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \leq \lambda\right) \leq \frac{(\lambda + c)^2}{V_n}.$$

■

Interpretation. Kolmogorov's maximal inequality gives an upper bound on the probability that the partial sums leave a prescribed interval, whereas the reverse inequality gives an upper bound on the probability that they remain inside that interval. In particular, if

$$V_n \rightarrow \infty,$$

then for every fixed $\lambda > 0$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \leq \lambda\right) \rightarrow 0.$$

1.3. Kolmogorov's Convergence Criterion

Theorem 1.3 (Kolmogorov's Convergence Criterion). *Let $X_1, X_2, \dots$ be independent random variables with*

$$\mathbb{E}[X_n] = 0, \quad \text{Var}(X_n) < \infty.$$

If

$$\sum_{n=1}^{\infty} \text{Var}(X_n) < \infty,$$

then

$$\sum_{n=1}^{\infty} X_n$$

converges almost surely.

Proof. Let

$$S_n = \sum_{k=1}^n X_k.$$

We show that $\{S_n\}$ is almost surely Cauchy.

For $m > n$, Kolmogorov's maximal inequality applied to

$$X_{n+1}, \dots, X_m$$

gives

$$\mathbb{P}\left(\max_{n < k \leq m} |S_k - S_n| > \varepsilon\right) \leq \frac{1}{\varepsilon^2} \sum_{j=n+1}^m \text{Var}(X_j).$$

Letting $m \rightarrow \infty$,

$$\mathbb{P}\left(\sup_{k > n} |S_k - S_n| > \varepsilon\right) \leq \frac{1}{\varepsilon^2} \sum_{j=n+1}^{\infty} \text{Var}(X_j).$$

Since

$$\sum_{j=1}^{\infty} \text{Var}(X_j) < \infty,$$

the right-hand side converges to zero.

To obtain an almost-sure statement, choose integers n_r such that

$$\sum_{j=n_r+1}^{\infty} \text{Var}(X_j) \leq 2^{-3r}.$$

Then

$$\mathbb{P}\left(\sup_{k > n_r} |S_k - S_{n_r}| > 2^{-r}\right) \leq 2^{2r} 2^{-3r} = 2^{-r}.$$

The probabilities are summable. Hence, by the Borel–Cantelli lemma, almost surely,

$$\sup_{k > n_r} |S_k - S_{n_r}| \leq 2^{-r}$$

for all sufficiently large r .

Thus $\{S_n\}$ is almost surely Cauchy, and therefore converges almost surely. ■

1.4. Kolmogorov's Three-Series Theorem

The convergence criterion requires finite variances and summability of those variances. The three-series theorem gives a necessary and sufficient condition for almost-sure convergence without assuming finite second moments.

Let $X_1, X_2, \dots$ be independent. Fix $A > 0$ and define the truncated variables

$$X_n^{(A)} = X_n \mathbf{1}\{|X_n| \leq A\}.$$

Theorem 1.4 (Kolmogorov's Three-Series Theorem). *The series*

$$\sum_{n=1}^{\infty} X_n$$

converges almost surely if and only if, for some $A > 0$, the following three series satisfy:

1.

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > A) < \infty;$$

2.

$$\sum_{n=1}^{\infty} \mathbb{E}[X_n^{(A)}] \text{ converges};$$

3.

$$\sum_{n=1}^{\infty} \text{Var}(X_n^{(A)}) < \infty.$$

The three conditions control three distinct phenomena:

$\sum \mathbb{P}(X_n > A) < \infty$: large jumps occur only finitely often, $\sum \mathbb{E}[X_n^{(A)}]$ converges : the deterministic drift converges, $\sum \text{Var}(X_n^{(A)}) < \infty$: the centered random fluctuations converge.

Indeed, by Borel–Cantelli, the first condition implies

$$X_n = X_n^{(A)}$$

eventually almost surely.

Sufficiency direction: Writing

$$X_n^{(A)} = (X_n^{(A)} - \mathbb{E}[X_n^{(A)}]) + \mathbb{E}[X_n^{(A)}],$$

the second condition controls the deterministic part, while the third condition, together with Kolmogorov's convergence criterion, controls the centered random part.

Necessity direction: Suppose

$$\sum_{n=1}^{\infty} X_n$$

converges almost surely. Fix $A > 0$ and define

$$Y_n = X_n \mathbf{1}\{|X_n| \leq A\}.$$

Since convergence of the series implies $X_n \rightarrow 0$ almost surely,

$$\mathbb{P}(|X_n| > A \text{ i.o.}) = 0.$$

By independence and the second Borel–Cantelli lemma,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > A) < \infty.$$

Consequently, $X_n = Y_n$ eventually almost surely, and hence

$$\sum_{n=1}^{\infty} Y_n$$

also converges almost surely.

Let $Y'_1, Y'_2, \dots$ be an independent copy of $Y_1, Y_2, \dots$, and define

$$Z_n = Y_n - Y'_n.$$

Then the Z_n are independent and centered,

$$|Z_n| \leq 2A, \quad \text{Var}(Z_n) = 2\text{Var}(Y_n),$$

and

$$\sum_{n=1}^{\infty} Z_n$$

converges almost surely.

We claim that

$$\sum_{n=1}^{\infty} \text{Var}(Z_n) < \infty.$$

Suppose, toward a contradiction, that

$$\sum_{n=1}^{\infty} \text{Var}(Z_n) = \infty.$$

Let

$$S_k = \sum_{j=1}^k Z_j.$$

By the reverse Kolmogorov maximal inequality, for every $M > 0$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| > M\right) \geq 1 - \frac{(M + 2A)^2}{\sum_{j=1}^n \text{Var}(Z_j)}.$$

Letting $n \rightarrow \infty$ gives

$$\mathbb{P}\left(\sup_{k \geq 1} |S_k| > M\right) = 1.$$

Since this holds for every positive integer M ,

$$\mathbb{P}\left(\sup_{k \geq 1} |S_k| = \infty\right) = 1.$$

But $\sum Z_n$ converges almost surely, so its sequence of partial sums must be bounded almost surely. This is a contradiction. Therefore

$$\sum_{n=1}^{\infty} \text{Var}(Z_n) < \infty.$$

Since

$$\text{Var}(Z_n) = 2\text{Var}(Y_n),$$

we obtain

$$\sum_{n=1}^{\infty} \text{Var}(Y_n) < \infty.$$

By Kolmogorov's convergence criterion,

$$\sum_{n=1}^{\infty} (Y_n - \mathbb{E}[Y_n])$$

converges almost surely. Since $\sum Y_n$ also converges almost surely, their difference converges, and therefore

$$\sum_{n=1}^{\infty} \mathbb{E}[Y_n]$$

converges.

Thus all three conditions are necessary:

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > A) < \infty, \quad \sum_{n=1}^{\infty} \mathbb{E}[Y_n] \text{ converges,} \quad \sum_{n=1}^{\infty} \text{Var}(Y_n) < \infty.$$

2. The Strong Law of Large Numbers

2.1. The Finite-Variance Case

We first prove the strong law under a finite-variance assumption.

Theorem 2.1 (Strong Law: Finite-Variance Case). *Let $X_1, X_2, \dots$ be i.i.d. with*

$$\mathbb{E}[X_1] = \mu, \quad \text{Var}(X_1) = \sigma^2 < \infty.$$

Then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu.$$

Proof. Define

$$Y_n = X_n - \mu.$$

Then

$$\mathbb{E}[Y_n] = 0, \quad \text{Var}(Y_n) = \sigma^2.$$

Consider the series

$$\sum_{n=1}^{\infty} \frac{Y_n}{n}.$$

The terms are independent and centered, and

$$\sum_{n=1}^{\infty} \text{Var}\left(\frac{Y_n}{n}\right) = \sigma^2 \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

By Kolmogorov's convergence criterion,

$$\sum_{n=1}^{\infty} \frac{Y_n}{n}$$

converges almost surely.

We now use Kronecker's lemma.

Lemma 2.2 (Kronecker). *If*

$$\sum_{n=1}^{\infty} \frac{a_n}{n}$$

converges, then

$$\frac{1}{n} \sum_{k=1}^n a_k \longrightarrow 0.$$

Applying the lemma pathwise with $a_n = Y_n$ gives

$$\frac{1}{n} \sum_{k=1}^n Y_k \xrightarrow{a.s.} 0.$$

Therefore

$$\bar{X}_n - \mu = \frac{1}{n} \sum_{k=1}^n (X_k - \mu) \xrightarrow{a.s.} 0.$$

Hence

$$\boxed{\bar{X}_n \xrightarrow{a.s.} \mu.}$$

■

2.2. The General i.i.d. Strong Law

The finite-variance assumption is unnecessary. For i.i.d. random variables, existence of the first absolute moment is enough.

Theorem 2.3 (Kolmogorov's Strong Law). *Let $X_1, X_2, \dots$ be i.i.d. and suppose*

$$\mathbb{E}|X_1| < \infty.$$

Then

$$\bar{X}_n \xrightarrow{a.s.} \mathbb{E}[X_1].$$

Proof. Write

$$\mu = \mathbb{E}[X_1].$$

Define the truncated variables

$$Y_n = X_n \mathbf{1}\{|X_n| \leq n\}.$$

We proceed in three steps.

Step 1: truncation changes only finitely many terms.

Since X_n are identically distributed,

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n).$$

Because $\mathbb{E}|X_1| < \infty$,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n) < \infty.$$

Hence Borel–Cantelli gives

$$\mathbb{P}(X_n \neq Y_n \text{ infinitely often}) = 0.$$

Thus

$$X_n = Y_n$$

eventually almost surely, and consequently

$$\frac{1}{n} \sum_{k=1}^n (X_k - Y_k) \xrightarrow{a.s.} 0.$$

Step 2: control the centered truncated variables.

Consider

$$\sum_{n=1}^{\infty} \frac{Y_n - \mathbb{E}[Y_n]}{n}.$$

The terms are independent and centered. We claim

$$\sum_{n=1}^{\infty} \frac{\text{Var}(Y_n)}{n^2} < \infty.$$

Since

$$\text{Var}(Y_n) \leq \mathbb{E}[Y_n^2],$$

it is enough to show

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \mathbb{E} [X_1^2 \mathbf{1}_{\{|X_1| \leq n\}}] < \infty.$$

By Tonelli's theorem,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \mathbb{E} [X_1^2 \mathbf{1}_{\{|X_1| \leq n\}}] = \mathbb{E} \left[X_1^2 \sum_{n \geq |X_1|} \frac{1}{n^2} \right].$$

For $x \geq 1$,

$$\sum_{n \geq x} \frac{1}{n^2} \leq \frac{C}{x}.$$

Therefore

$$X_1^2 \sum_{n \geq |X_1|} \frac{1}{n^2} \leq C|X_1|$$

up to an irrelevant bounded contribution when $|X_1| < 1$. Since $\mathbb{E}|X_1| < \infty$,

$$\sum_{n=1}^{\infty} \frac{\text{Var}(Y_n)}{n^2} < \infty.$$

Kolmogorov's convergence criterion therefore gives

$$\sum_{n=1}^{\infty} \frac{Y_n - \mathbb{E}[Y_n]}{n} \text{ converges almost surely.}$$

By Kronecker's lemma,

$$\frac{1}{n} \sum_{k=1}^n (Y_k - \mathbb{E}[Y_k]) \xrightarrow{a.s.} 0.$$

Step 3: identify the deterministic limit.

Since

$$Y_n = X_1 \mathbf{1}\{|X_1| \leq n\} \quad \text{in distribution,}$$

dominated convergence gives

$$\mathbb{E}[Y_n] = \mathbb{E}[X_1 \mathbf{1}\{|X_1| \leq n\}] \longrightarrow \mathbb{E}[X_1] = \mu.$$

Hence, by Cesàro convergence,

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E}[Y_k] \longrightarrow \mu.$$

Combining the three steps,

$$\begin{aligned} \frac{1}{n} \sum_{k=1}^n X_k &= \frac{1}{n} \sum_{k=1}^n (X_k - Y_k) + \frac{1}{n} \sum_{k=1}^n (Y_k - \mathbb{E}[Y_k]) + \frac{1}{n} \sum_{k=1}^n \mathbb{E}[Y_k] \\ &\xrightarrow{a.s.} 0 + 0 + \mu. \end{aligned}$$

Therefore

$$\boxed{\bar{X}_n \xrightarrow{a.s.} \mathbb{E}[X_1].}$$

■